

Theoretical studies of excitonic transitions in two dimensional materials under photoexcitation of scalar and vector vortex beams

1. Introduction
 2. Spin-orbital angular momentum interaction in light
 3. Light-matter interaction
 4. Exciton wave packet
- Summary and outlook

1. Introduction

2. Spin-orbital angular momentum interaction in light

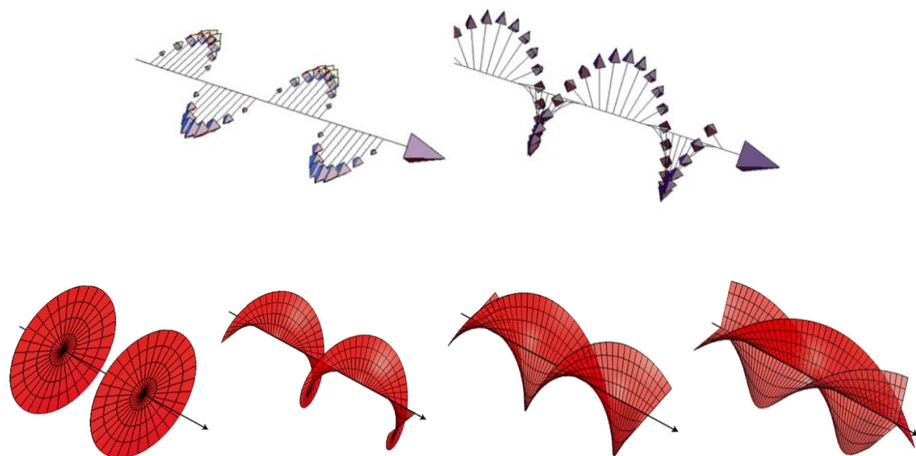
3. Light-matter interaction

4. Exciton wave packet

Summary and outlook

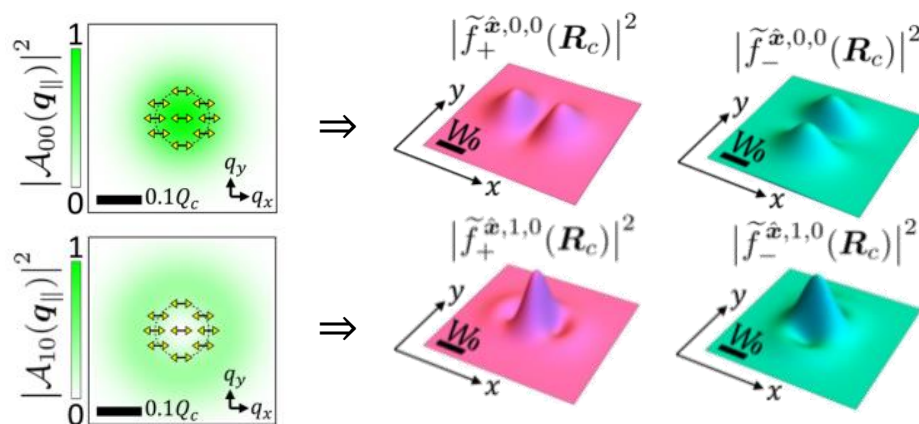
1. Introduction

- Angular momentum in light

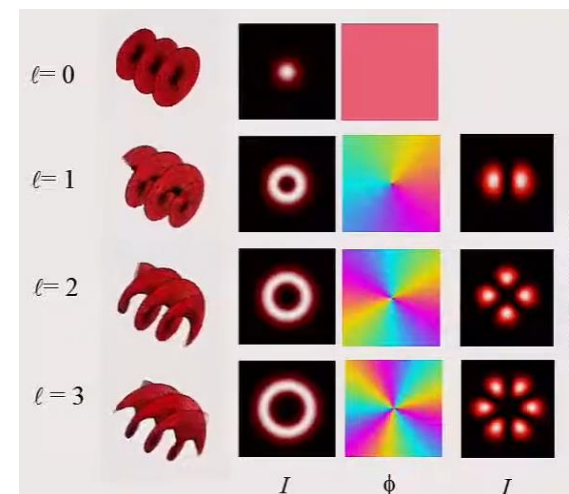


- Polarization of light / Helical phase fronts for $\ell = 0, 1, 2, 3$
A. M. Yao & M. J. Padgett, *Adv. Opt. Photonics* **3**, 161 (2011)

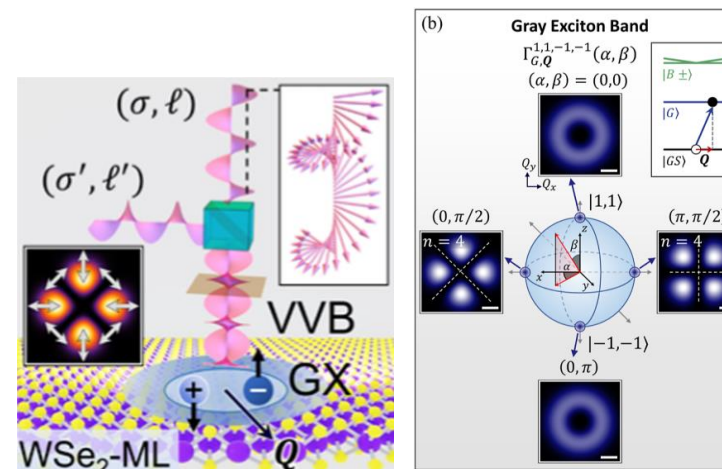
- Light matter interaction



- OAM can influence exciton's wave packet structures
G.-H. Peng *et al.*, *Nano Lett.* **19**, 2299 (2019)



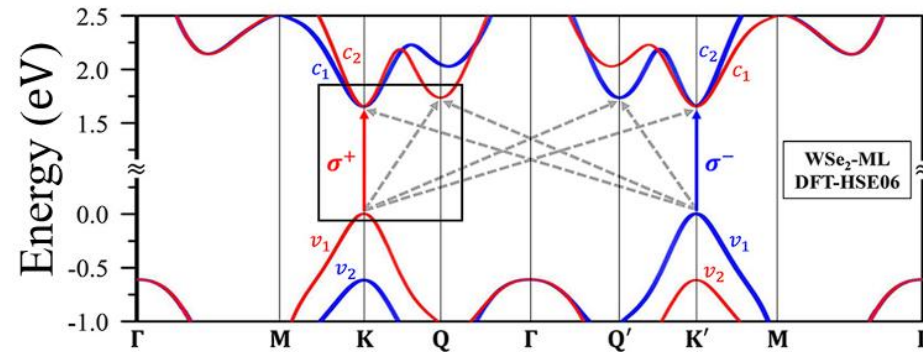
- LG beams and vector vortex beams (transverse plane)
Reproduced from Miles J. Padgett, International School on Parametric Nonlinear Optics (2015)



- SOI influence optical transition rate through vector vortex beam
O. J. G. Sanchez, G.-H. Peng *et al.*, *ACS Nano* **18**, 11425 (2024)

1. Introduction

- Polarization of light
 - Optical spin



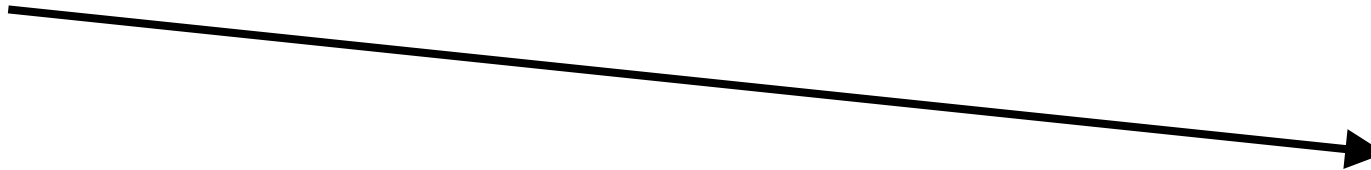
• WSe₂, calculated band structure
G.-H. Peng *et al.*, *Nano Lett.* **19**, 2299 (2019)

- Spin of electron

- Exciton with spin
- Bright*2/dark/gray

- dipole

- OAM



- COM of exciton

1. Introduction

1. Introduction

2. Spin-orbital angular momentum interaction (SOI) in light

- Laguerre-Gaussian beams
- Longitudinal component : SOI effect
- Longitudinal component : Vector vortex beams

3. Light-matter interaction

4. Exciton wave packet

Summary and outlook

2. Laguerre-Gaussian (LG) beam : Derivation

LG beam is solved as vector potential $\mathbf{A}(\mathbf{r})$ from paraxial Helmholtz equation within **Lorentz gauge**.

Maxwell equations

scalar Helmholtz equation

$$(\nabla^2 + q_0^2)A(\mathbf{r}) = 0$$

paraxial approximation

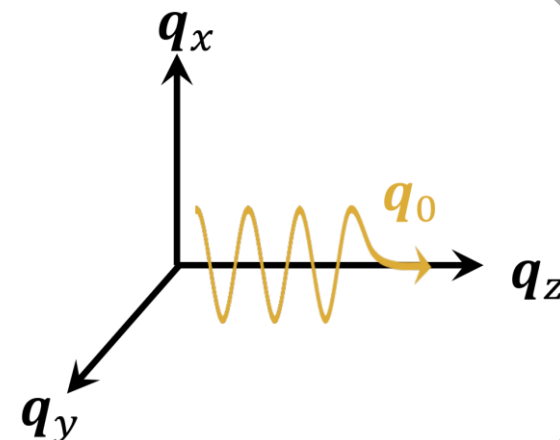
$$\left(\nabla_{\parallel}^2 + \frac{\partial^2}{\partial z^2} + 2iq_0 \frac{\partial}{\partial z}\right)U(\mathbf{r}) = 0$$

Ansatz

Paraxial wave

$$A(\mathbf{r}) = U(\mathbf{r})e^{iq_0 z}$$

wave vector $\mathbf{q}_z \approx \mathbf{q}_0$



$$\left|\frac{\partial^2}{\partial z^2}U(\mathbf{r})\right| \ll q_0 \left|\frac{\partial}{\partial z}U(\mathbf{r})\right|$$

Approx.

$$\left|\frac{\partial^2}{\partial z^2}U(\mathbf{r})\right| \ll |\nabla_{\parallel}^2 U(\mathbf{r})|$$

paraxial Helmholtz equation

$$\left(\nabla_{\parallel}^2 + 2iq_0 \frac{\partial}{\partial z}\right)U(\mathbf{r}) \approx 0$$

Fail

Coulomb gauge condition

$$\nabla \cdot \mathbf{A}^C(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} \cdot \nabla_{\parallel} A^C(\mathbf{r})$$

$$= \left(\frac{\partial}{\partial x} + i\sigma \frac{\partial}{\partial y}\right)A^C(\mathbf{r}) = 0, \quad \text{for } \hat{\mathbf{e}}_{\sigma} = \hat{\mathbf{x}} + i\sigma \hat{\mathbf{y}} \\ \sigma = \pm 1$$

eventually constraint the transverse component

$$\nabla_{\parallel}^2 A^C(\mathbf{r}) = \left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)A^C(\mathbf{r}) = 0.$$

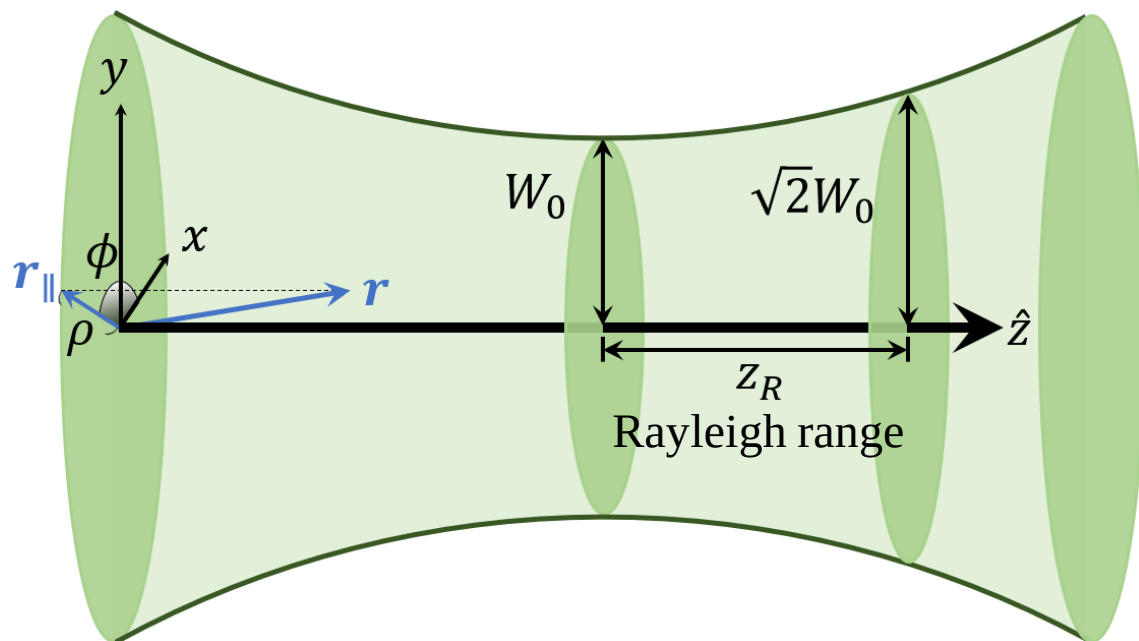
2. Laguerre-Gaussian (LG) beam : Distribution

LG beam is solved as vector potential $\mathbf{A}(\mathbf{r})$ from paraxial Helmholtz equation within Lorentz gauge.

With **long Rayleigh range approximation** :

$$A_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell| + p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$



- Polarization unit vector :

$$\hat{\mathbf{e}}_{\sigma} = \hat{\mathbf{x}} + i\sigma\hat{\mathbf{y}}, \quad \sigma = \pm 1$$

(LG beam carries SAM of $\sigma\hbar$)

- Topological charge:

$$\ell \in \mathbb{Z}$$

(LG beam carries OAM of $\ell\hbar$)

- Beam waist : W_0
- Rayleigh range : $z_R = \frac{1}{2} q_0 W_0^2$
- Radial mode index: integer $p \geq 0$
- Constant amplitude : A_0^{LG}
- Generalized Laguerre polynomial :

$$L_p^{|\ell|}(x) = \sum_{m=0}^p (-1)^m \frac{(|\ell|+p)!}{(p-m)! (|\ell|+m)! m!} x^m$$

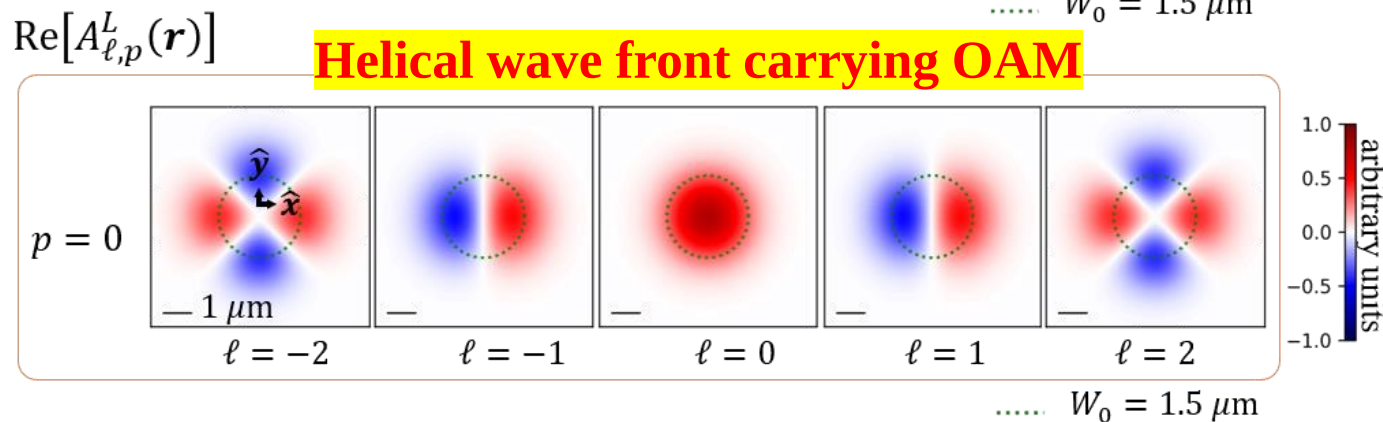
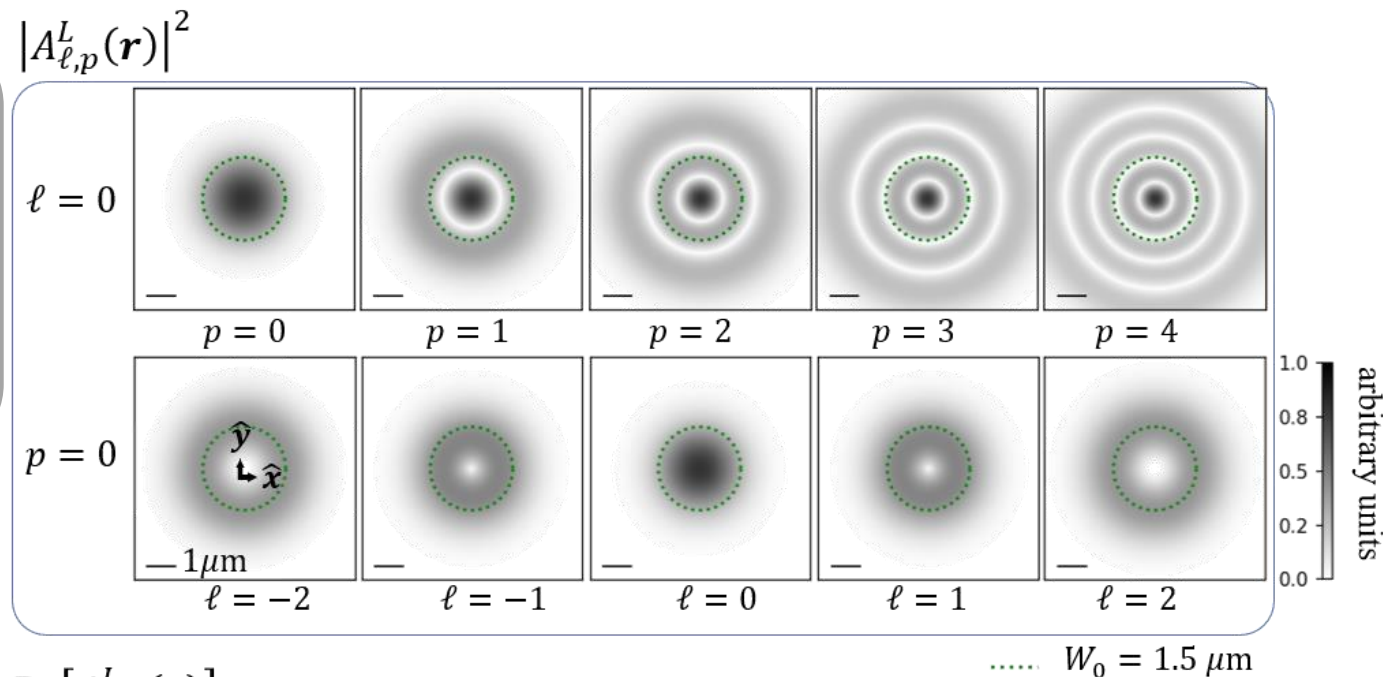
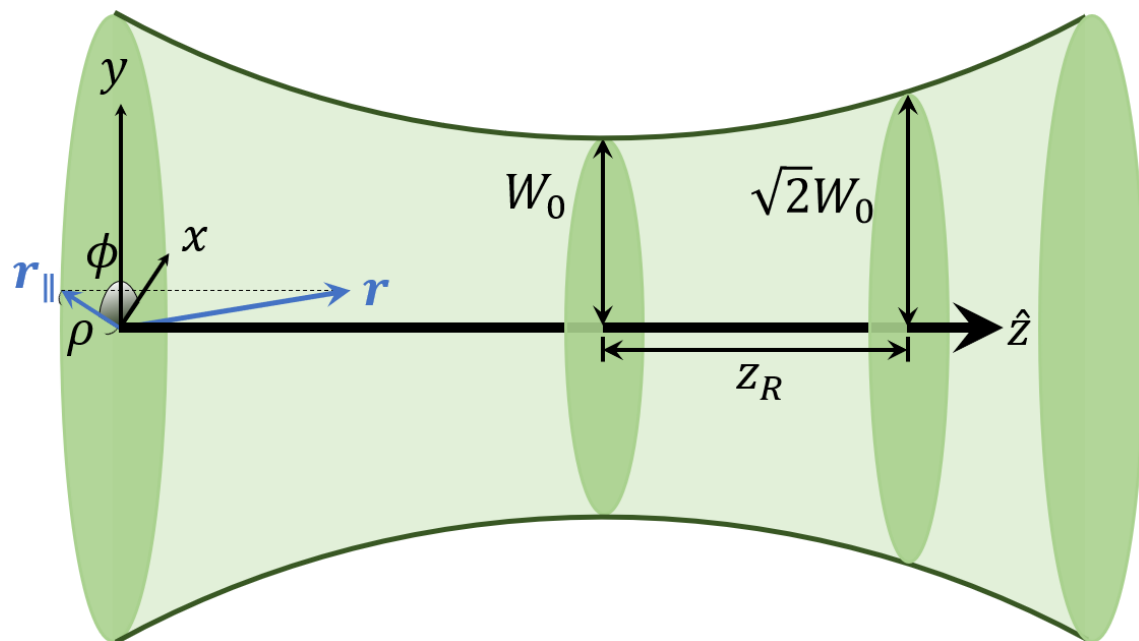
2. Laguerre-Gaussian (LG) beam : Distribution

LG beam is solved as vector potential $\mathbf{A}(\mathbf{r})$ from paraxial Helmholtz equation within Lorentz gauge.

With **long Rayleigh range approximation** :

$$A_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

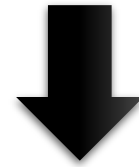
$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell| + p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$



2. Laguerre-Gaussian (LG) beam : Requirement

LG beam in real space / Lorentz gauge :

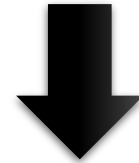
$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r})$$



LG beam in angular spectrum / Coulomb gauge :

$$\mathcal{A}_{\sigma,\ell,p}^C(q_{\parallel})$$

What we really need



Theory of light matter interaction

$$\Gamma^{S,Q} = \frac{2\pi}{\hbar} |\tilde{M}^{S,Q}|^2 \rho^L(\hbar\omega = E^{S,Q})$$

$$\tilde{M}^{S,Q} \approx \frac{E_g}{2i\hbar} \delta_{q_{\parallel}Q} \mathcal{A}^C(q_{\parallel}) \cdot D^{S,Q*}$$

2. Laguerre-Gaussian (LG) beam : Angular spectrum

LG beam in Lorentz gauge / **real space** :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell| + p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

2. Laguerre-Gaussian (LG) beam : Angular spectrum

LG beam in Lorentz gauge / **real space** :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

LG beam in Lorentz gauge / **angular spectrum** :

$$\mathcal{A}_{\sigma,\ell,p}^L(\mathbf{q}_{\parallel}) = \hat{\mathbf{e}}_{\sigma} \mathcal{A}_{\ell,p}^L(\mathbf{q}_{\parallel}) \approx \hat{\mathbf{e}}_{\sigma} \tilde{F}_{|\ell|,p}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}}$$

$$\tilde{F}_{|\ell|,p}(q_{\parallel}) = \frac{2\pi(-i)^{|\ell|}}{\Omega} \mathcal{H}_{|\ell|}[f_{\ell,p}(\rho)]$$

Hankel transform of order $|\ell|$

Method

Expanded **in plane-wave basis**

$$\mathbf{A}(\mathbf{r}) = \sum_{\mathbf{q}} \mathcal{A}(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}}$$

Define Fourier transformation

$$f_{\ell,p}(\rho) e^{i\ell\phi} = u_{\ell,p}(\mathbf{r}_{\parallel}) = \frac{1}{(2\pi)^2} \int d^2 \mathbf{q}_{\parallel} F_{\ell,p}^{\text{LG}}(q_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}}$$

$$F_{\ell,p}^{\text{LG}}(q_{\parallel}) = \int d^2 \mathbf{r}_{\parallel} u_{\ell,p}(\mathbf{r}_{\parallel}) e^{-i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}}$$

Results

$$\mathcal{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \sum_{\mathbf{q}_{\parallel}} \mathcal{A}_{\sigma,\ell,p}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z}$$

2. Laguerre-Gaussian (LG) beam : Angular spectrum

LG beam in Lorentz gauge / **real space** :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

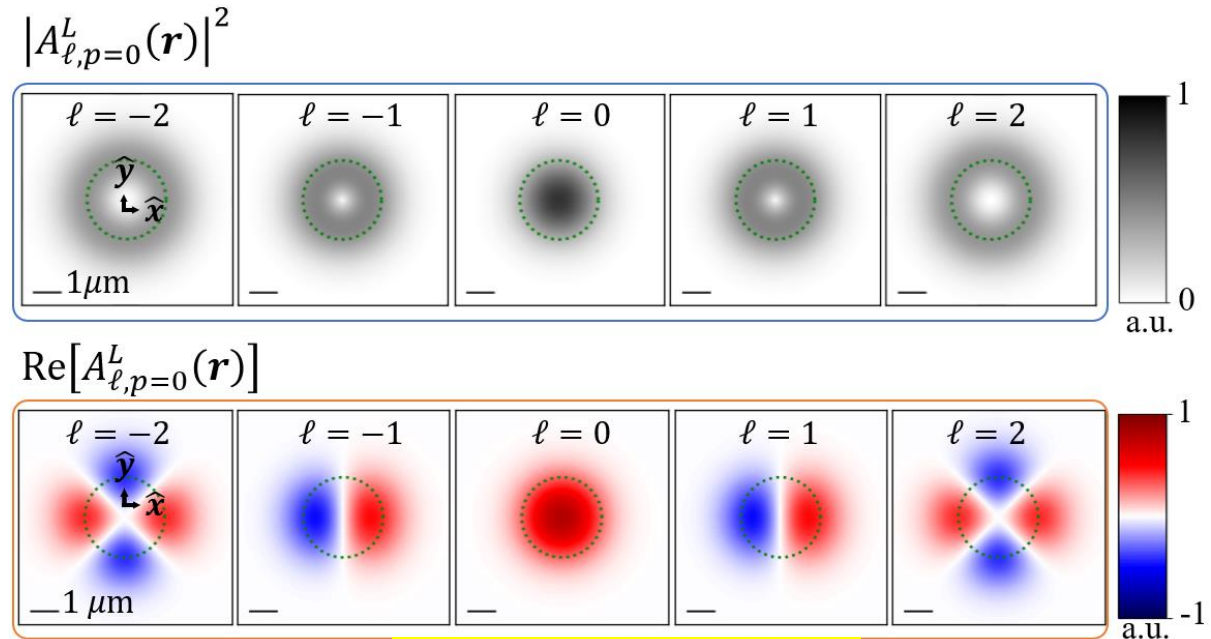
$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

LG beam in Lorentz gauge / **angular spectrum** :

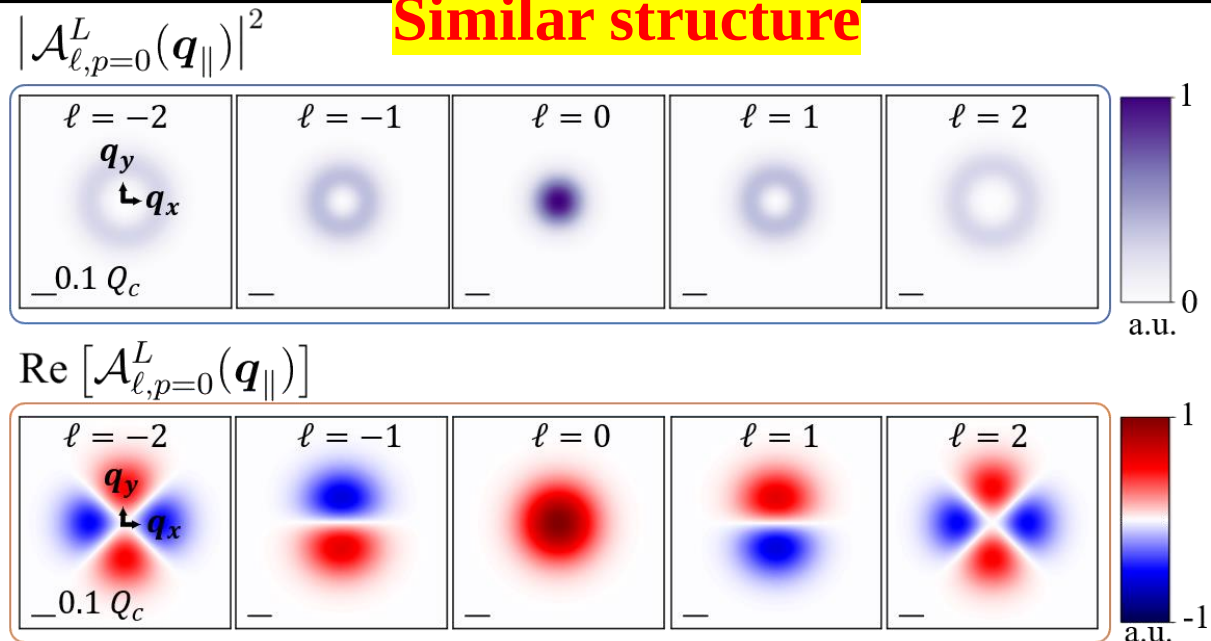
$$\mathcal{A}_{\sigma,\ell,p}^L(q_{\parallel}) = \hat{\mathbf{e}}_{\sigma} \mathcal{A}_{\ell,p}^L(q_{\parallel}) \approx \hat{\mathbf{e}}_{\sigma} \tilde{F}_{|\ell|,p}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}}$$

$$\tilde{F}_{|\ell|,p}(q_{\parallel}) = \frac{2\pi(-i)^{|\ell|}}{\Omega} \mathcal{H}_{|\ell|}[f_{\ell,p}(\rho)]$$


 Hankel transform of order $|\ell|$



Similar structure



2. Laguerre-Gaussian (LG) beam : Coulomb gauge

LG beam in **Lorentz gauge** / real space :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\boldsymbol{\varepsilon}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\boldsymbol{\varepsilon}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell| + p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

2. Laguerre-Gaussian (LG) beam : Coulomb gauge

LG beam in **Lorentz gauge** / real space :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

LG beam in **Coulomb gauge** / real space:

$$\mathbf{A}_{\sigma,\ell,p}(r) \equiv \mathbf{A}_{\sigma,\ell,p}^C(r) \approx \tilde{A}_{\ell,p}^{\parallel}(r) \hat{\mathbf{e}}_{\sigma} + \tilde{A}_{\sigma,\ell,p}^z(r) \hat{\mathbf{z}}$$

Transverse component :

$$\tilde{A}_{\ell,p}^{\parallel}(r) \approx f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

Longitudinal component :

$$\tilde{A}_{\sigma,\ell,p}^z(r) \approx \frac{i}{\sqrt{2}q_0\rho} \left[(|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2} \right] f_{\ell,p}(\rho) e^{i(\sigma+\ell)\phi} e^{iq_0 z}$$

Additional component

Method

Relation from Maxwell's equations

$$\mathbf{A}_{\sigma,\ell,p}^C(r) = \mathbf{A}_{\sigma,\ell,p}^L(r) + \frac{\nabla \left(\nabla \cdot \mathbf{A}_{\sigma,\ell,p}^L(r) \right)}{q_0^2}$$

2. Laguerre-Gaussian (LG) beam : Coulomb gauge

LG beam in **Lorentz gauge** / real space :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell| + p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

LG beam in **Coulomb gauge** / real space:

$$\mathbf{A}_{\sigma,\ell,p}(r) \equiv \mathbf{A}_{\sigma,\ell,p}^C(r) \approx \tilde{A}_{\ell,p}^{\parallel}(r) \hat{\mathbf{e}}_{\sigma} + \tilde{A}_{\sigma,\ell,p}^z(r) \hat{\mathbf{z}}$$

Transverse component :

$$\tilde{A}_{\ell,p}^{\parallel}(r) \approx f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

Longitudinal component :

$$\tilde{A}_{\sigma,\ell,p}^z(r) \approx \frac{i}{\sqrt{2}q_0\rho} \left[(|\ell| + \text{SOI} \sigma \ell) - \frac{2\rho^2}{W_0^2} \right] f_{\ell,p}(\rho) e^{i(\sigma+\ell)\phi} e^{iq_0 z}$$

2. Laguerre-Gaussian (LG) beam : SOI appear

LG beam in **Lorentz gauge** / real space :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

$$f_{\ell,p}(\rho) = A_0^{\text{LG}} \sqrt{\frac{2p!}{\pi(|\ell|+p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} e^{-\frac{\rho^2}{W_0^2}}$$

— $|\tilde{A}_{\ell,p=0}^{\parallel}(r)|$
— $|\tilde{A}_{\sigma=1,\ell,p=0}^z(r)|$
— $|\tilde{A}_{\sigma=-1,\ell,p=0}^z(r)|$

	Transverse	Longitudinal
$\pm\ell$	symmetric	Asymmetric SOI
$W_0 \downarrow$	merely changed	Larger

LG beam in **Coulomb gauge** / real space:

$$\mathbf{A}_{\sigma,\ell,p}(r) \equiv \mathbf{A}_{\sigma,\ell,p}^C(r) \approx \tilde{A}_{\ell,p}^{\parallel}(r) \hat{\mathbf{e}}_{\sigma} + \tilde{A}_{\sigma,\ell,p}^z(r) \hat{\mathbf{z}}$$

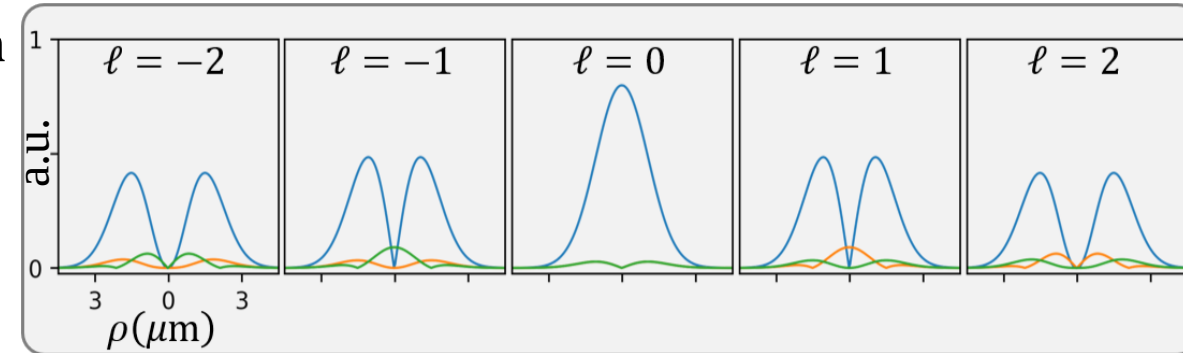
Transverse component :

$$\tilde{A}_{\ell,p}^{\parallel}(r) \approx f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

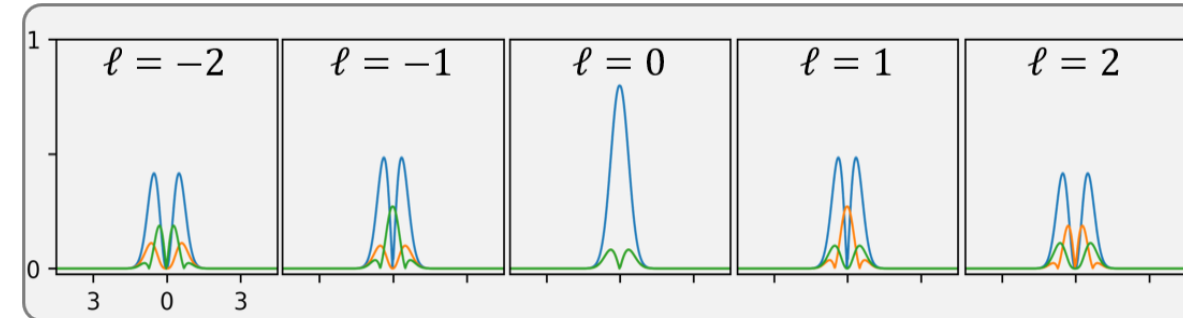
Longitudinal component :

$$\tilde{A}_{\sigma,\ell,p}^z(r) \approx \frac{i}{\sqrt{2}q_0\rho} \left[(|\ell| + \text{SOI}) - \frac{2\rho^2}{W_0^2} \right] f_{\ell,p}(\rho) e^{i(\sigma+\ell)\phi} e^{iq_0 z}$$

$W_0 = 1.5\mu\text{m}$



$W_0 = 0.5\mu\text{m}$



2. Longitudinal component : SOI effect

LG beam solutions:

Real / Lorentz :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

Angular / Lorentz :

$$\mathcal{A}_{\sigma,\ell,p}^L(q_{\parallel}) = \hat{\mathbf{e}}_{\sigma} \mathcal{A}_{\ell,p}^L(q_{\parallel}) \approx \hat{\mathbf{e}}_{\sigma} \tilde{F}_{|\ell|,p}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}}$$

Real / Coulomb :

$$\mathbf{A}_{\sigma,\ell,p}(r) \equiv \mathbf{A}_{\sigma,\ell,p}^C(r) \approx \tilde{A}_{\ell,p}^{\parallel}(r) \hat{\mathbf{e}}_{\sigma} + \tilde{A}_{\sigma,\ell,p}^z(r) \hat{\mathbf{z}}$$

2. Longitudinal component : SOI effect

LG beam solutions:

Real / Lorentz :

$$\mathbf{A}_{\sigma,\ell,p}^L(\mathbf{r}) = \hat{\mathbf{e}}_{\sigma} A_{\ell,p}^L(\mathbf{r}) \approx \hat{\mathbf{e}}_{\sigma} f_{\ell,p}(\rho) e^{i\ell\phi} e^{iq_0 z}$$

Angular / Lorentz :

$$\mathcal{A}_{\sigma,\ell,p}^L(\mathbf{q}_{\parallel}) = \hat{\mathbf{e}}_{\sigma} \mathcal{A}_{\ell,p}^L(\mathbf{q}_{\parallel}) \approx \hat{\mathbf{e}}_{\sigma} \tilde{F}_{|\ell|,p}(\mathbf{q}_{\parallel}) e^{i\ell\phi_{\mathbf{q}_{\parallel}}}$$

Real / Coulomb :

$$\mathbf{A}_{\sigma,\ell,p}(\mathbf{r}) \equiv \mathbf{A}_{\sigma,\ell,p}^C(\mathbf{r}) \approx \tilde{A}_{\ell,p}^{\parallel}(\mathbf{r}) \hat{\mathbf{e}}_{\sigma} + \tilde{A}_{\sigma,\ell,p}^z(\mathbf{r}) \hat{\mathbf{z}}$$

$$\mathcal{A}_{\sigma,\ell,p}^C(\mathbf{q}_{\parallel}) \quad \text{What we really need}$$

LG beam in **angular spectrum** / **Coulomb gauge** :

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) \equiv \mathcal{A}_{\sigma,\ell,p=0}^C(\mathbf{q}_{\parallel}) \approx \tilde{\mathcal{A}}_{\ell,p}^{\parallel}(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{\sigma} + \tilde{\mathcal{A}}_{\sigma,\ell,p}^z(\mathbf{q}_{\parallel}) \hat{\mathbf{z}}$$

Transverse component :

$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\ell,p=0}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) e^{i\ell\phi_{\mathbf{q}_{\parallel}}}$$

Longitudinal component :

$$\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\sigma,\ell,p=0}^z(\mathbf{q}_{\parallel}) \approx -\frac{\sin \theta_{\mathbf{q}}}{\sqrt{2}} \tilde{F}_{|\ell|}(\mathbf{q}_{\parallel}) e^{i(\sigma+\ell)\phi_{\mathbf{q}_{\parallel}}} \quad \text{SOI}$$

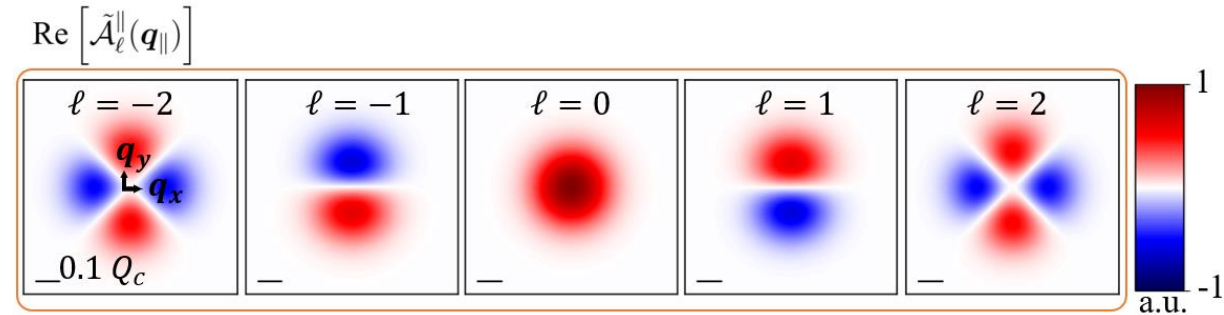
Method

Expanded in plane-wave basis

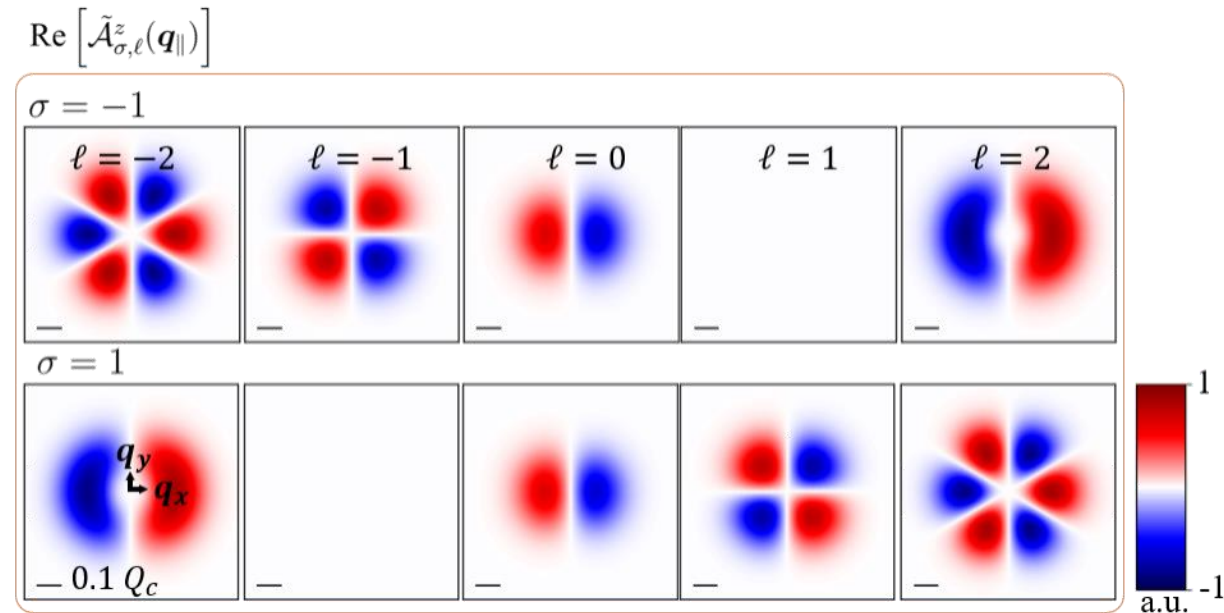
$$\mathbf{A}_{\sigma,\ell,p}^C(\mathbf{r}) = \frac{1}{(2\pi)^2} \int d\mathbf{q}_{\parallel} \mathcal{A}_{\sigma,\ell,p}^C(\mathbf{q}_{\parallel}) e^{i\mathbf{q} \cdot \mathbf{r}}$$

2. Longitudinal component : SOI effect

Transverse component :



Longitudinal component :



$$\mathcal{A}_{\sigma,\ell,p}^c(\mathbf{q}_{\parallel})$$

LG beam in **angular spectrum** / **Coulomb gauge** :

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) \equiv \mathcal{A}_{\sigma,\ell,p=0}^c(\mathbf{q}_{\parallel}) \approx \tilde{\mathcal{A}}_{\ell,p}^{\parallel}(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{\sigma} + \tilde{\mathcal{A}}_{\sigma,\ell,p}^z(\mathbf{q}_{\parallel}) \hat{\mathbf{z}}$$

Transverse component :

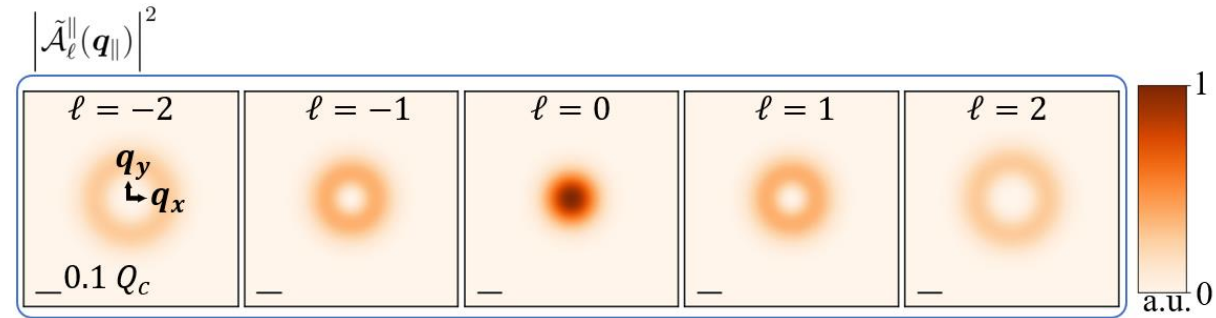
$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\ell,p=0}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\ell\phi_{\mathbf{q}_{\parallel}}}$$

Longitudinal component :

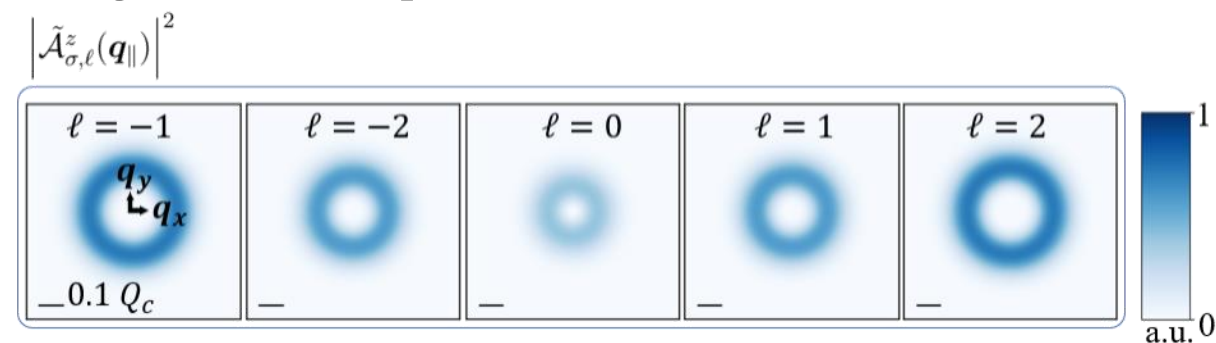
$$\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\sigma,\ell,p=0}^z(\mathbf{q}_{\parallel}) \approx -\frac{\sin \theta_{\mathbf{q}}}{\sqrt{2}} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i(\sigma+\ell)\phi_{\mathbf{q}_{\parallel}}} \text{SOI}$$

2. Longitudinal component : SOI effect

Transverse component :



Longitudinal component :



$$\mathcal{A}_{\sigma,\ell,p}^C(\mathbf{q}_{\parallel})$$

LG beam in **angular spectrum** / **Coulomb gauge** :

$$\mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) \equiv \mathcal{A}_{\sigma,\ell,p=0}^C(\mathbf{q}_{\parallel}) \approx \tilde{\mathcal{A}}_{\ell,p}^{\parallel}(\mathbf{q}_{\parallel}) \hat{\mathbf{e}}_{\sigma} + \tilde{\mathcal{A}}_{\sigma,\ell,p}^z(\mathbf{q}_{\parallel}) \hat{\mathbf{z}}$$

Transverse component :

$$\tilde{\mathcal{A}}_{\ell}^{\parallel}(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\ell,p=0}^{\parallel}(\mathbf{q}_{\parallel}) \approx \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\ell\phi_{\mathbf{q}_{\parallel}}}$$

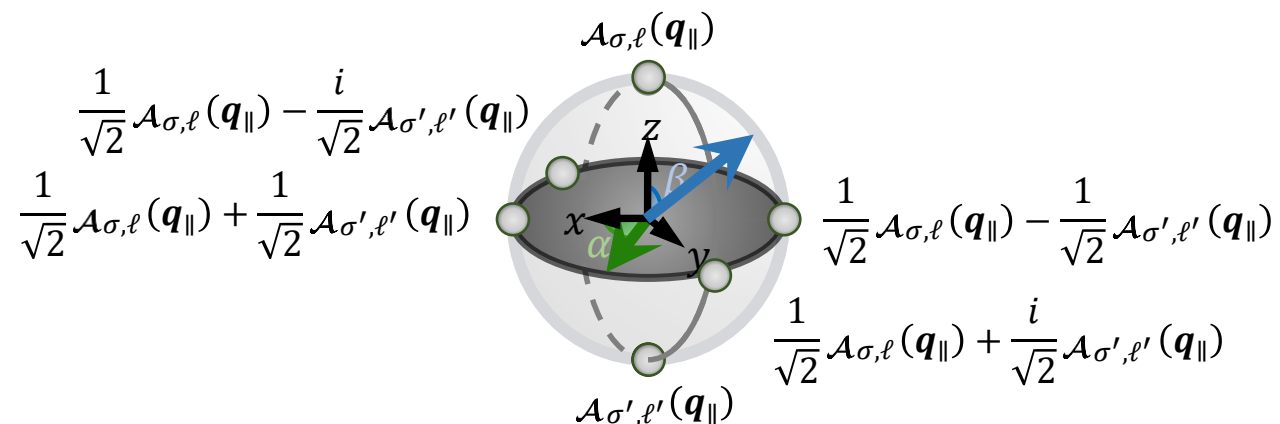
Longitudinal component :

$$\tilde{\mathcal{A}}_{\sigma,\ell}^z(\mathbf{q}_{\parallel}) \equiv \tilde{\mathcal{A}}_{\sigma,\ell,p=0}^z(\mathbf{q}_{\parallel}) \approx -\frac{\sin \theta_{\mathbf{q}}}{\sqrt{2}} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i(\sigma+\ell)\phi_{\mathbf{q}_{\parallel}}} \text{SOI}$$

2. Vector vortex beams (VVB)

Forming vector vortex beams (VVB) by Superimposing LG beams

$$\begin{aligned}\mathcal{A}_{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) &= \cos\left(\frac{\beta}{2}\right) \mathcal{A}_{\sigma,\ell}(\mathbf{q}_{\parallel}) + e^{i\alpha} \sin\left(\frac{\beta}{2}\right) \mathcal{A}_{\sigma',\ell'}(\mathbf{q}_{\parallel}) \\ &= \mathcal{A}_{\sigma,\ell,\sigma',\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta) + \mathcal{A}_{\sigma,\ell,\sigma',\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta)\end{aligned}$$



- Poincaré sphere for VVB states

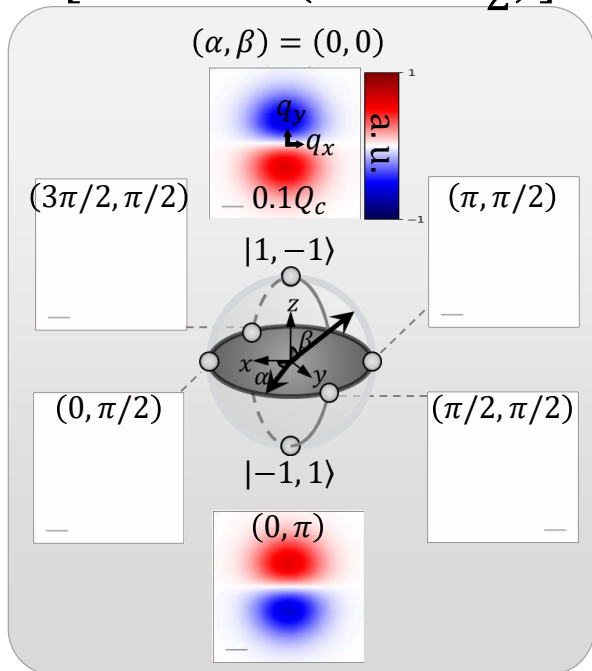
2. Transverse component : VVB

If $\tilde{F}_{|\ell|}(q_{\parallel}) = \tilde{F}_{|\ell'|}(q_{\parallel})$

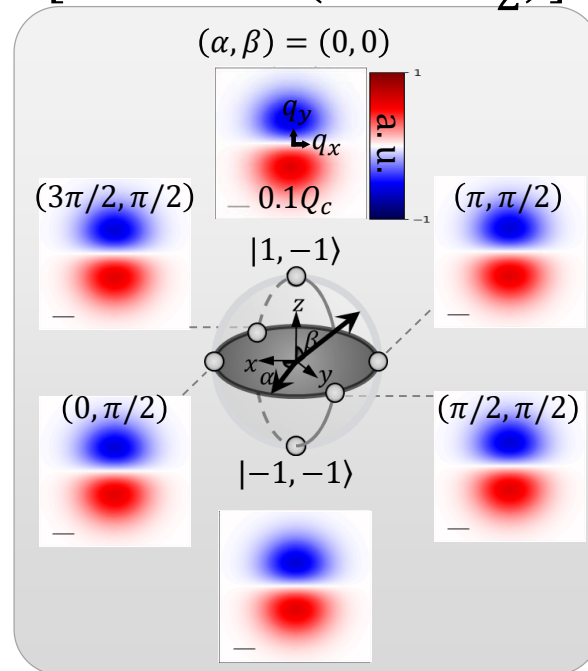
$$\mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \equiv \mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) \hat{\varepsilon}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha)$$

$$\mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) = \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\bar{\ell}\phi_{q_{\parallel}}}$$

$$\text{Re} \left[\mathcal{A}_{1,-1,-1,1}^{\parallel}(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) \right]$$



$$\text{Re} \left[\mathcal{A}_{1,-1,-1,-1}^{\parallel}(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) \right]$$



Carrying phase : $\bar{\ell}\phi_{q_{\parallel}}$

$$\bar{\ell} \equiv \frac{\ell' + \ell}{2}$$

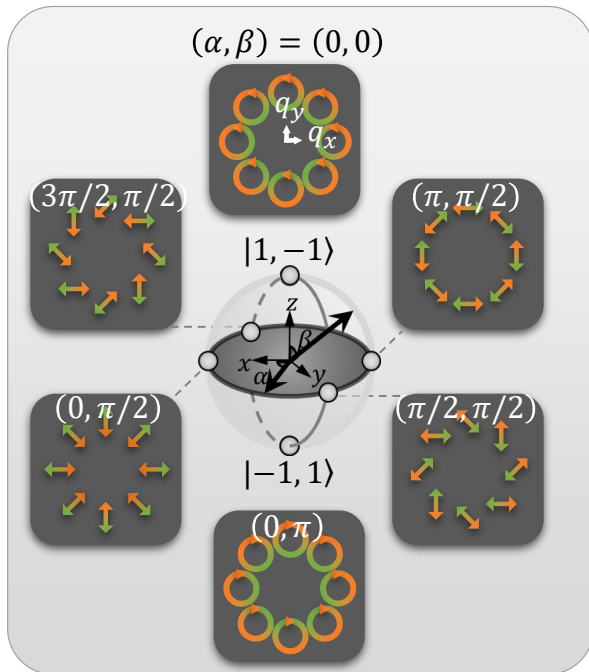
2. Transverse component : VVB

If $\tilde{F}_{|\ell|}(q_{\parallel}) = \tilde{F}_{|\ell'|}(q_{\parallel})$

$$\mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \equiv \mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) \hat{\mathbf{e}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha)$$

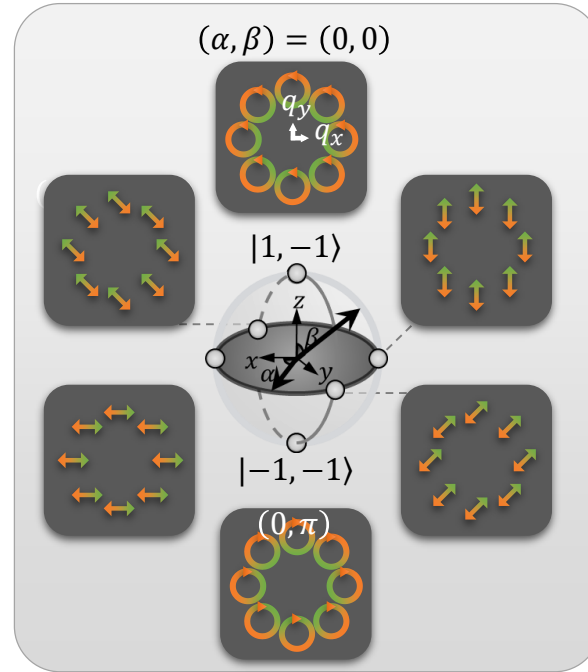
$$\mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) = \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\bar{\ell}\phi_{q_{\parallel}}} \quad \hat{\mathbf{e}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha) = e^{i\frac{\alpha}{2}} \left\{ \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{x}} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{y}} \right\}$$

$\text{Re}[\hat{\mathbf{e}}_{1,-1,-1,1}^{\parallel}(\mathbf{q}_{\parallel}; \alpha)]$



$$\Rightarrow \Delta\ell = 2$$

$\text{Re}[\hat{\mathbf{e}}_{1,-1,-1,-1}^{\parallel}(\mathbf{q}_{\parallel}; \alpha)]$



$$\Rightarrow \Delta\ell = 0$$

Carrying $q_{\parallel}$ dependence : $\frac{\Delta\ell}{2}\phi_{q_{\parallel}}$

$$\Delta\ell \equiv \ell' - \ell$$

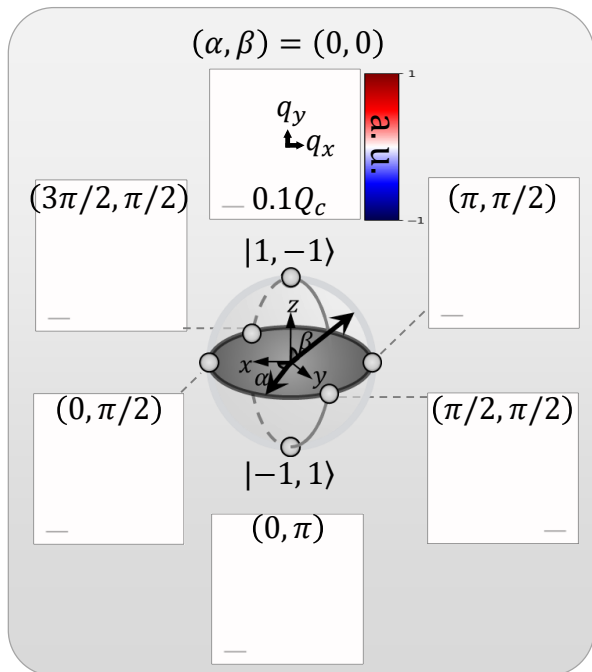
2. Longitudinal component : VVB

If $\tilde{F}_{|\ell|}(q_{\parallel}) = \tilde{F}_{|\ell'|}(q_{\parallel})$

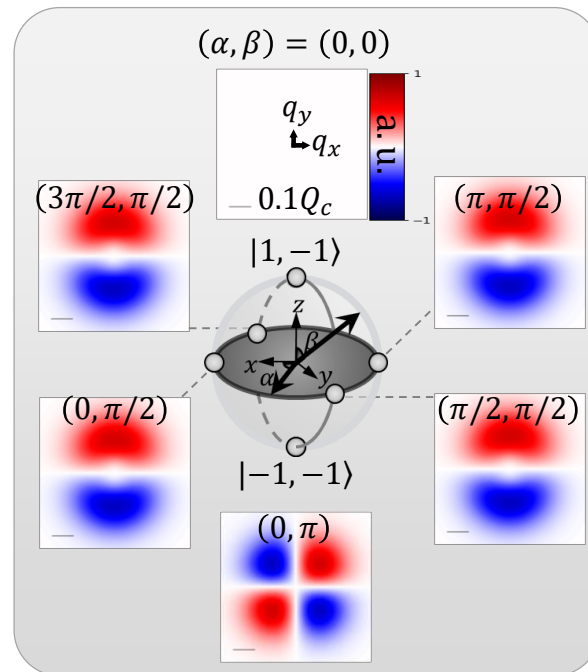
$$\mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \equiv \mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) \hat{\varepsilon}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha)$$

$$\mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\bar{J}\phi_{q_{\parallel}}}$$

$\text{Re} [\mathcal{A}_{1,-1,-1,1}^z(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2})]$



$\text{Re} [\mathcal{A}_{1,-1,-1,-1}^z(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2})]$



Carrying phase : $\bar{J}\phi_{q_{\parallel}}$

$$\bar{J} \equiv \frac{J' + J}{2}$$

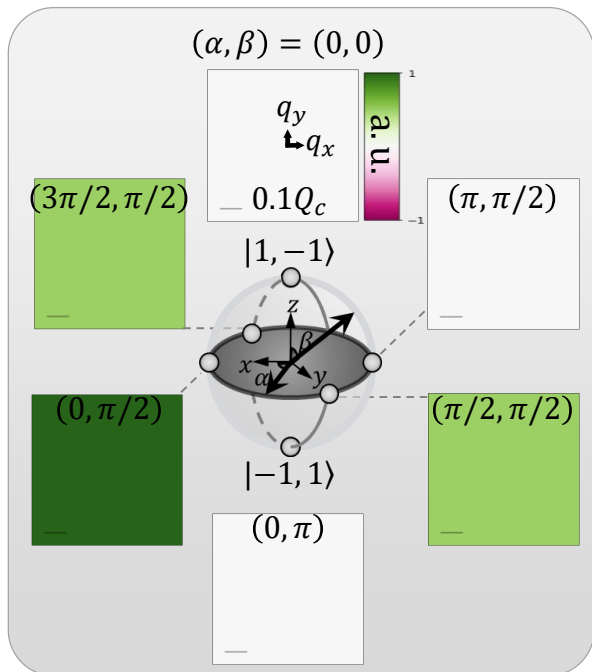
2. Longitudinal component : VVB

If $\tilde{F}_{|\ell|}(q_{\parallel}) = \tilde{F}_{|\ell'|}(q_{\parallel})$

$$\mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \equiv \mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) \hat{\mathbf{e}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha)$$

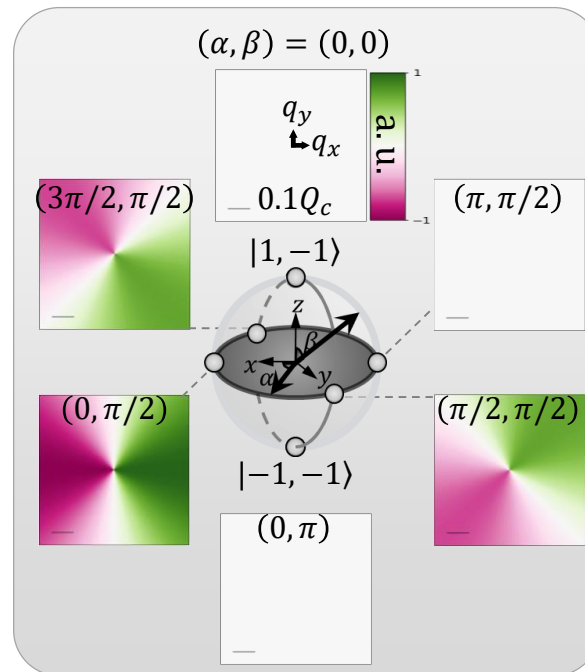
$$\mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\bar{J}\phi_{q_{\parallel}}} \quad \hat{\mathbf{e}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha) = e^{i\frac{\alpha}{2} \cos\left(\frac{\Delta J}{2} \phi_{q_{\parallel}} + \frac{\alpha}{2}\right)} \hat{\mathbf{z}}$$

$\text{Re}[\hat{\mathbf{e}}_{1,-1,-1,1}^z(\mathbf{q}_{\parallel}; \alpha)]$



$$\Rightarrow \Delta J = 0$$

$\text{Re}[\hat{\mathbf{e}}_{1,-1,-1,-1}^z(\mathbf{q}_{\parallel}; \alpha)]$



$$\Rightarrow \Delta J = -2$$

Carrying $q_{\parallel}$ dependence : $\frac{\Delta J}{2} \phi_{q_{\parallel}}$

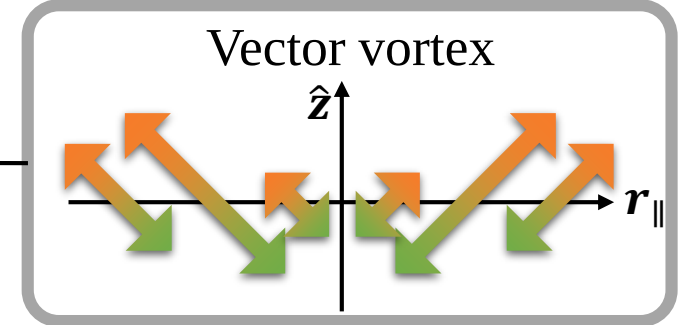
$$\Delta J \equiv J' - J$$

2. Brief overview : VVB

Summary for $\beta = \frac{\pi}{2}$, $\sigma = -\sigma' = 1$, and $\tilde{F}_{|\ell|}(q_{\parallel}) = \tilde{F}_{|\ell'|}(q_{\parallel})$

$$\mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \equiv \mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) \hat{\mathbf{e}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha)$$

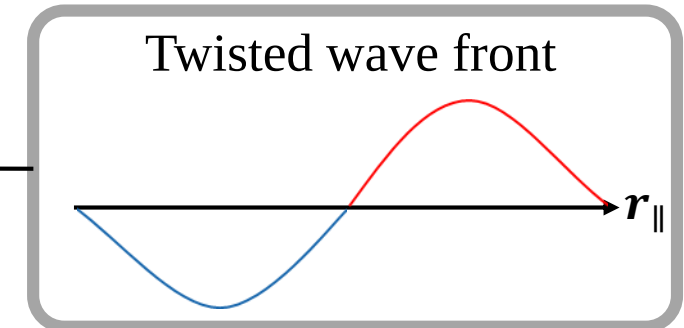
$$\mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \equiv \mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) \hat{\mathbf{e}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha)$$



$$\mathcal{A}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) = \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\bar{\ell}\phi_{q_{\parallel}}}$$

$$\mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \beta = \frac{\pi}{2}) = -\frac{\sin \theta_q}{2} \tilde{F}_{|\ell|}(q_{\parallel}) e^{i\bar{J}\phi_{q_{\parallel}}}$$

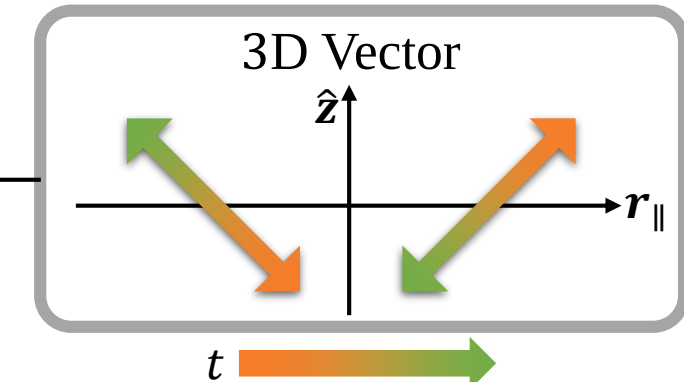
Average of AMs



$$\hat{\mathbf{e}}_{1,\ell,-1,\ell'}^{\parallel}(\mathbf{q}_{\parallel}; \alpha) = e^{i\frac{\alpha}{2}} \left\{ \cos\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{x}} + \sin\left(\frac{\Delta\ell}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{y}} \right\}$$

$$\hat{\mathbf{e}}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha) = e^{i\frac{\alpha}{2}} \cos\left(\frac{\Delta J}{2}\phi_{q_{\parallel}} + \frac{\alpha}{2}\right) \hat{\mathbf{z}}$$

Difference of AMs



2. Longitudinal component : VVB

What if $\tilde{F}_{|\ell|}(q_{\parallel}) \approx \tilde{F}_{|\ell'|}(q_{\parallel})$

$$\mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_{\mathbf{q}}}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\bar{\ell}| + \frac{\alpha}{2})} \left\{ \cos \left[\frac{1}{2} \left(\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin \left[\frac{1}{2} \left(\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] H_{|\ell|,|\ell'|}(q_{\parallel}) \right\} \hat{\mathbf{z}}$$

Dominant part $G_{|\ell|,|\ell'|}(q_{\parallel}) \equiv [F_{|\ell|}(q_{\parallel}) + F_{|\ell'|}(q_{\parallel})] \gg H_{|\ell|,|\ell'|}(q_{\parallel}) \equiv [F_{|\ell|}(q_{\parallel}) - F_{|\ell'|}(q_{\parallel})]$

Define

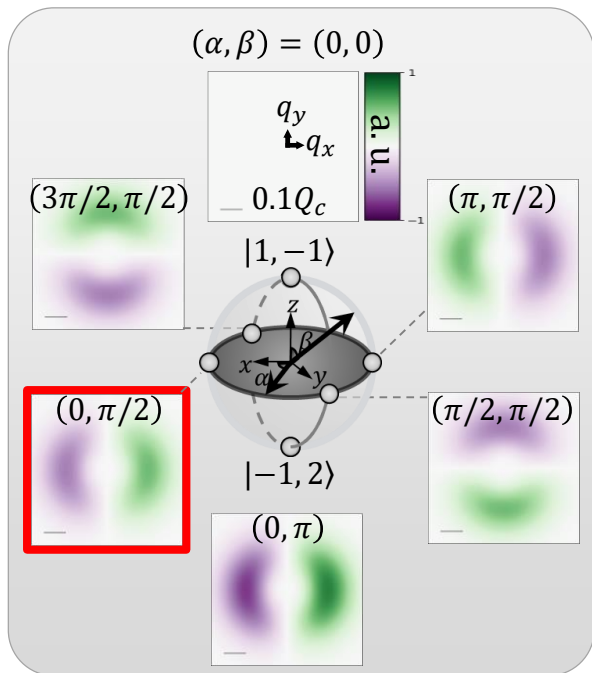
$$\begin{aligned} \bar{|\ell|} &\equiv (|\ell'| + |\ell|)/2 \\ \Delta |\ell| &\equiv |\ell'| - |\ell| \end{aligned}$$

2. Longitudinal component : VVB

What if $\tilde{F}_{|\ell|}(q_{\parallel}) \approx \tilde{F}_{|\ell'|}(q_{\parallel})$

$$\mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) = -\frac{\sin \theta_{\mathbf{q}}}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\ell| + \frac{\alpha}{2})} \left\{ \cos \left[\frac{1}{2} \left(\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin \left[\frac{1}{2} \left(\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] H_{|\ell|,|\ell'|}(q_{\parallel}) \right\} \hat{\mathbf{z}}$$

$$\text{Re} \left[\mathcal{A}_{1,-1,-1,2}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \right]$$



For $\alpha = 0$

$$\Delta J = 1 \Rightarrow \cos \left[\frac{1}{2} \phi_{q_{\parallel}} - \frac{\pi}{4} \right] \quad \text{Maximum at } \phi_{q_{\parallel}} = \frac{\pi}{2}$$

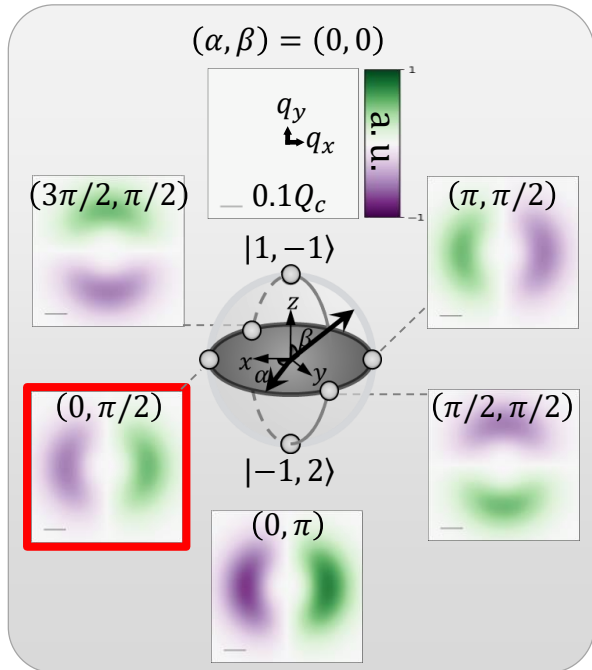
$$\bar{J} = \frac{1}{2} \Rightarrow \cos \left[\frac{1}{2} \phi_{q_{\parallel}} - \frac{3\pi}{4} \right] \quad \text{Maximum at } \phi_{q_{\parallel}} = \frac{3\pi}{2}$$

2. Longitudinal component : VVB

What if $\tilde{F}_{|\ell|}(q_{\parallel}) \approx \tilde{F}_{|\ell'|}(q_{\parallel})$

$$\mathcal{A}_{1,\ell,-1,\ell'}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) e^{-i\omega t} = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\ell| + \frac{\alpha}{2} - \omega t)} \left\{ \cos \left[\frac{1}{2} \left(\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin \left[\frac{1}{2} \left(\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] H_{|\ell|,|\ell'|}(q_{\parallel}) \right\} \hat{\mathbf{z}}$$

$$\text{Re} \left[\mathcal{A}_{1,-1,-1,2}^z(\mathbf{q}_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \right]$$



For $\alpha = 0$

$$\Delta J = 1 \Rightarrow \cos \left[\frac{1}{2} \phi_{q_{\parallel}} - \frac{\pi}{4} \right] \quad \text{Maximum at } \phi_{q_{\parallel}} = \frac{\pi}{2}$$

$$\bar{J} = \frac{1}{2} \Rightarrow \cos \left[\frac{1}{2} \phi_{q_{\parallel}} - \frac{3\pi}{4} - \omega t \right] \quad \text{Maximum at } \phi_{q_{\parallel}} = \frac{3\pi}{2} + 2\omega t$$

Can evolve with time

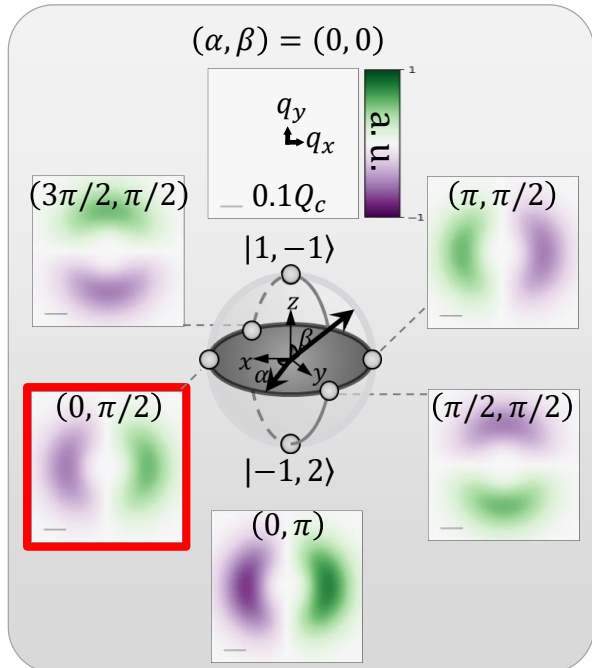
2. Longitudinal component : VVB

What if $\tilde{F}_{|\ell|}(q_{\parallel}) \approx \tilde{F}_{|\ell'|}(q_{\parallel})$

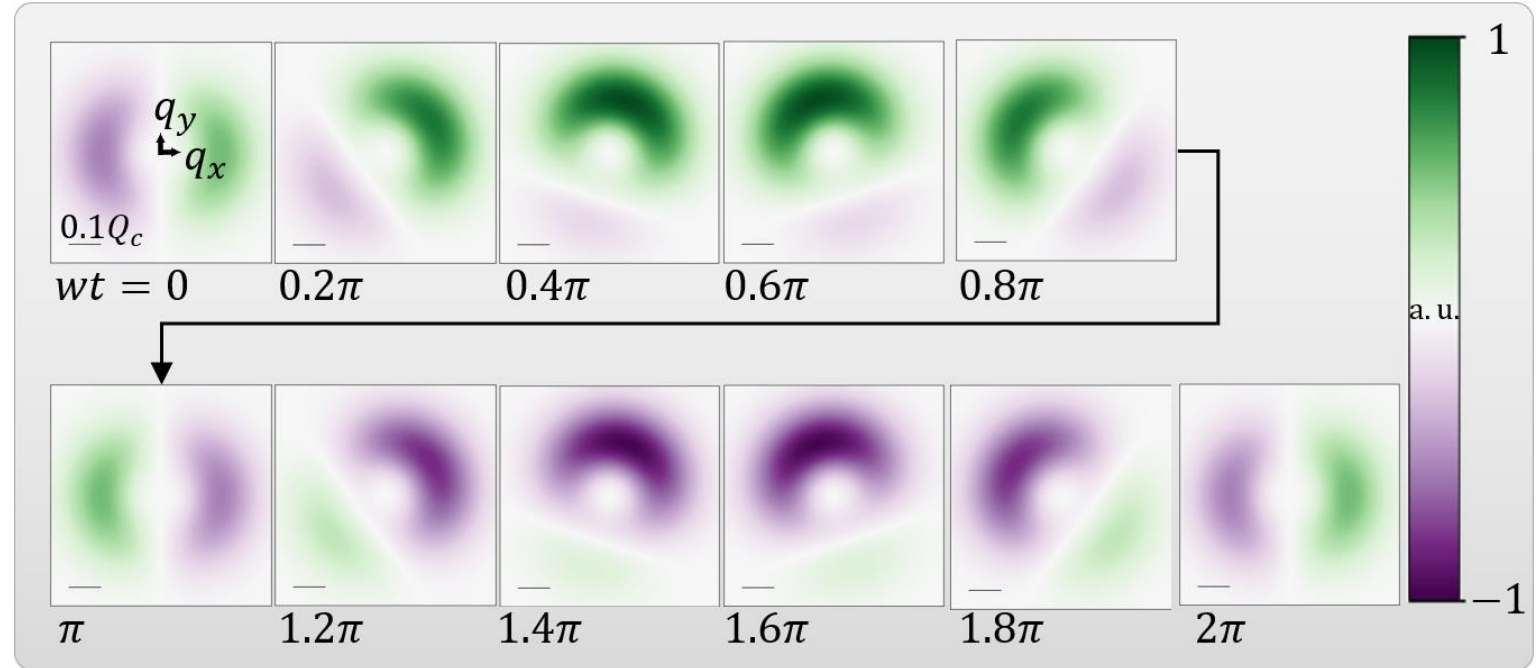
$$\mathcal{A}_{1,\ell,-1,\ell'}^z(q_{\parallel}; \alpha, \beta = \frac{\pi}{2}) e^{-i\omega t} = -\frac{\sin \theta_q}{2} e^{i(\bar{J}\phi_{q_{\parallel}} - \frac{\pi}{2}|\ell| + \frac{\alpha}{2} - \omega t)} \left\{ \cos \left[\frac{1}{2} \left(\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] G_{|\ell|,|\ell'|}(q_{\parallel}) - i \sin \left[\frac{1}{2} \left(\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} \Delta |\ell| + \alpha \right) \right] H_{|\ell|,|\ell'|}(q_{\parallel}) \right\} \hat{z}$$

Maximum at $\phi_{q_{\parallel}} = \frac{\pi}{2}$

$$\text{Re} \left[\mathcal{A}_{1,-1,-1,2}^z(q_{\parallel}; \alpha, \beta = \frac{\pi}{2}) \right]$$



$$\text{Re} \left[\mathcal{A}_{1,-1,-1,2}^z(q_{\parallel}; \alpha, \beta = \frac{\pi}{2}) e^{-i\omega t} \right]$$



1. Introduction

2. Spin-orbital angular momentum interaction in light

3. Light-matter interaction

- Selective excitation
- Encoding SOI information

4. Exciton wave packet

Summary and outlook

3. Light-matter interaction

1. Introduction
2. Spin-orbital angular momentum interaction in light
3. Light-matter interaction
- 4. Exciton wave packet**

Summary and outlook

4. Exciton wave packet

- Summary and outlook